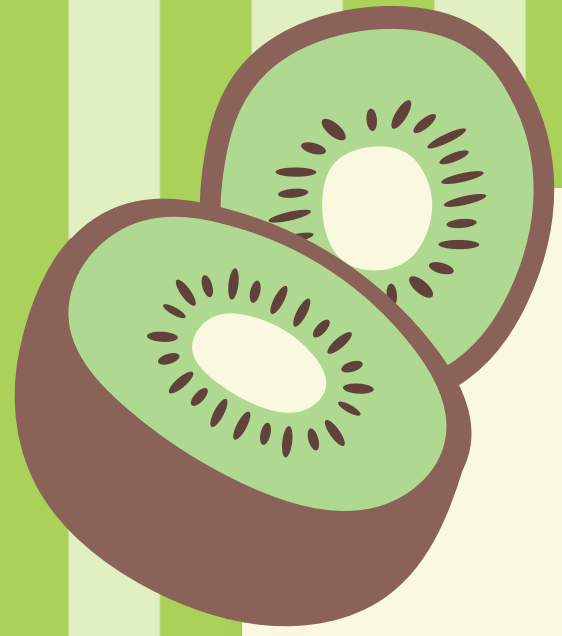




Fruit Freshness / Ripeness

Group 1

Nawaf - Riyadh - Mohammed - Norah - Rahaf



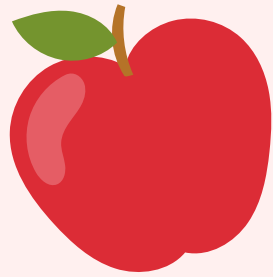
Objectives

- Detect fruit ripeness automatically using computer vision.
- Train a YOLOv8 Object Detection model.
- Deploy the model for real-time predictions.

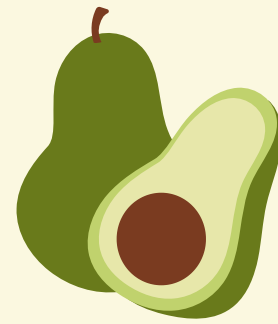


Problem

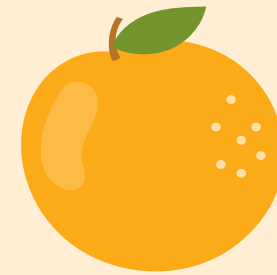
Many fruits look very similar during different ripeness stages and manual inspection:



Takes time



Can be inconsistent



**Depends on human
judgment**

Our goal was to classify fruit ripeness automatically from images.



Classes & Dataset



Classes

Apple Ripe - Apple Unripe - Banana Ripe - Banana Unripe -
Mango Ripe - Mango Overripe

Dataset

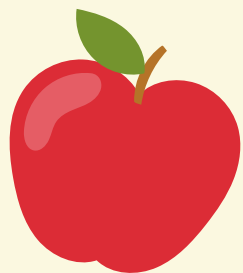
Source: Roboflow

Format: YOLOv8 OBB

Total Images: 813 images

Training / Validation / Test split

592 images / 148 images 73 images



Model Performance

We trained 3 different models

Model	Epochs	Image Size	mAP50-95
Run 1	25	512	0.9205
Run 2	50	512	0.9237
Run 3	25	640	0.9194

Selected the model with the highest mAP50-95

Final Precision: 99.24%

Final Recall: 98.17%

Final mAP50: 98.59%

Final mAP50-95: 92.37%

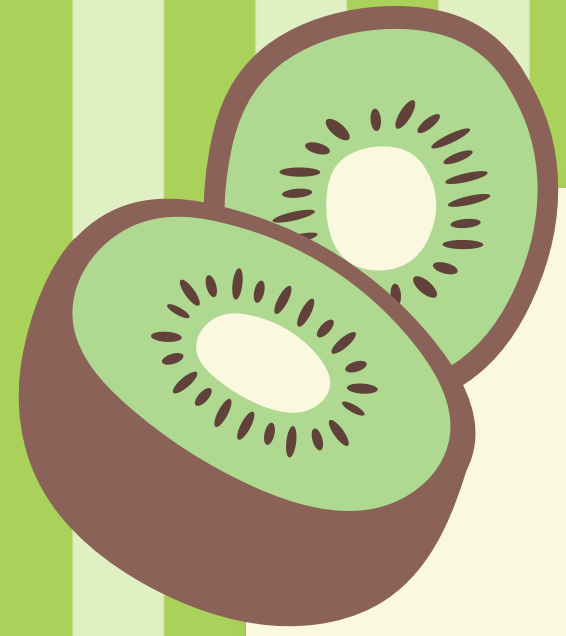
What Went Wrong

Similar ripeness stages
looked almost identical

Some classes had fewer images

Lighting and camera angles
affected predictions

A few fruits were confused with
neighboring ripeness stages



Thank You

Group 1

Nawaf - Riyadh - Mohammed - Norah - Rahaf